S2 Coursework: Coal Mining Accident Analysis

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1. Accident Data

The total number of accidents recorded is $N = 191$ over a period of $L = 40550$ days. The mean rate of accidents is 0.00471 accidents/day or 1.72 accidents/year. The cumulative plot shows non linear increase which suggests the accident rate may not be constant.

$$\bar{\lambda} = \frac{N}{L} \approx 0.0047 \text{ accidents/day.}$$

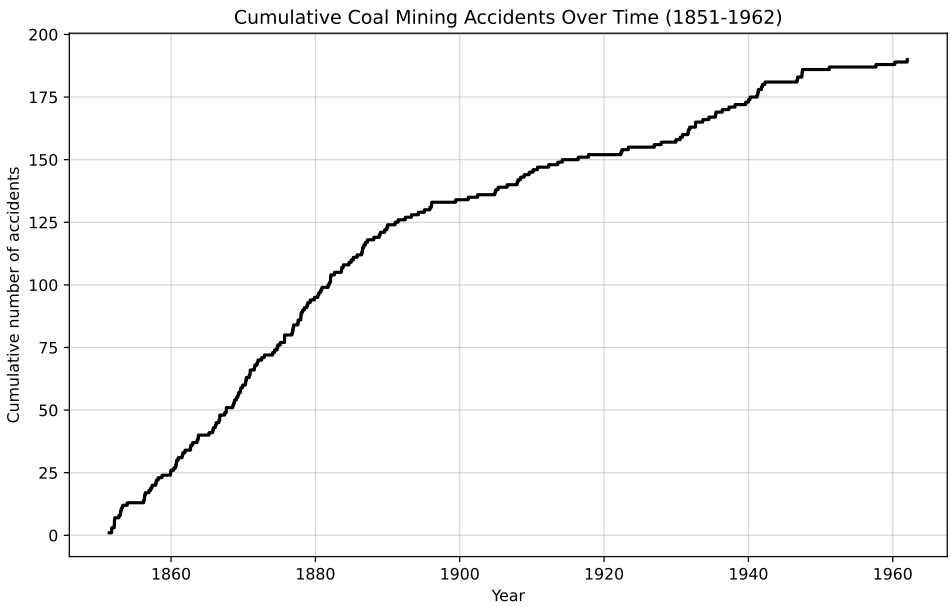

Figure 1: Cumulative number of coal mining accidents from 1851 to 1962. There were 191 accidents recorded over 40550 days.

2. Priors

(a) Qualitative Comparison

The plain order statistics prior involves drawing k change points from a uniform distribution over the interval $[0, L]$ and then sorting them. This means that the points are independent before sorting and can be clustered tightly together. As shown in Figure 2, the marginal distributions are broad and can peak directly at the boundaries.

The even-numbered prior is constructed by drawing $2k + 1$ points uniformly, sorting them, and then taking the evenly indexed points. This results in a repulsive effect between points, encouraging more evenly spaced change points. It also has the effect of preventing peaks at the

boundaries, which is appropriate for a change point, as the first and last points are always odd and therefore always excluded. In Figure 2 we can also see that the marginal distributions are more distinctly separated and narrower than those of the plain order priors.

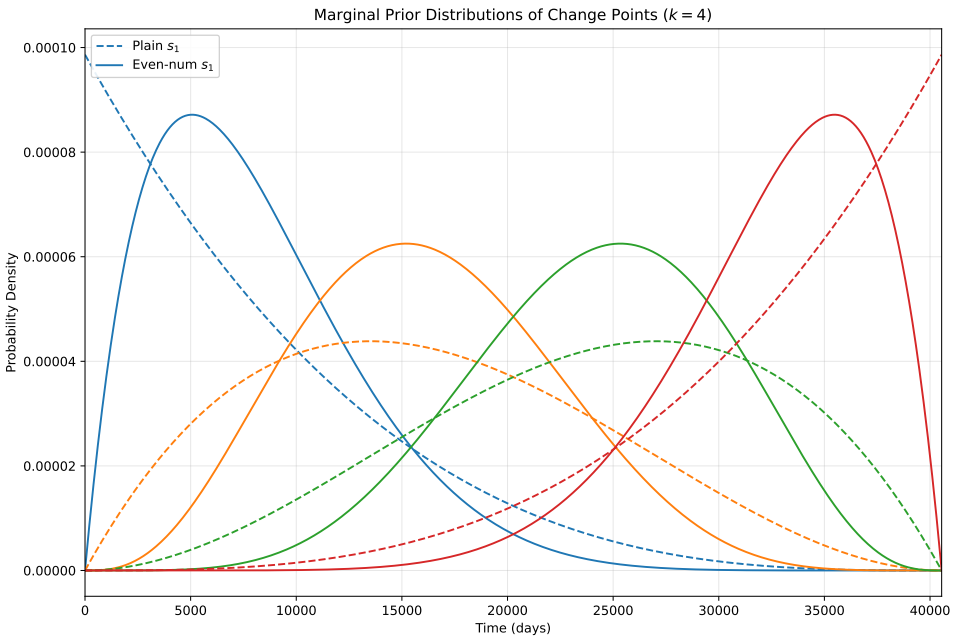

Figure 2: Comparison of plain and even-numbered order statistics priors for $k = 4$ across different values of j. Plain order priors are shown with a dashed line and even-numbered priors with a solid line.

(b) PDF Derivation

To derive the even-numbered pdf we start with the fact that the plain ordered pdf for $2k+1$ points on the unit interval $[0, 1]$ is a constant

$$P(u_{(1)}, u_{(2)}, \ldots, u_{(2k+1)}) = (2k+1)!$$

This is valid only on the strictly ordered region $0 < u_{(1)} < u_{(2)} < \cdots < u_{(2k+1)} < 1$.

For the even-numbered pdf we only keep even-numbered order statistics $\hat{s}_j = u_{(2j)}$ so must marginalise the plain order pdf over all the other discarded points. We know that the points are ordered and on the unit interval $[0, 1]$ so the integral is

$$P(\hat{s}_1, \ldots, \hat{s}_k) = \int_0^{\hat{s}_1} \int_{\hat{s}_1}^{\hat{s}_2} \cdots \int_{\hat{s}_k}^{1} (2k+1)!\, du_{(2k+1)} \ldots du_{(3)} du_{(1)}$$

Because the integrand $(2k+1)!$ is a constant, each integral becomes the length of the interval between the 2 points on the bounds, or the gap $\hat{g}_j$ between them.

$$P(\hat{s}_1, \ldots, \hat{s}_k) = (2k+1)!(\hat{s}_1)(\hat{s}_2 - \hat{s}_1) \ldots (1 - \hat{s}_k)$$

$$= (2k+1)! \prod_{j=1}^{k+1} \hat{g}_j$$

We now need to map this from the unit interval $[0, 1]$ to the actual timeline $[0, L]$. We are given the scaling relationship $s_j = L\hat{s}_j$ so can scale the gaps according to $\hat{g}_j = g_j/L$ and substitute this into the equation

$$P(\hat{s}_1, \ldots, \hat{s}_k) = (2k+1)! \prod_{j=1}^{k+1} \left(\frac{g_j}{L}\right)$$

Since L is a constant it can be pulled out of the product

$$P(\hat{s}_1, \ldots, \hat{s}_k) = \frac{(2k+1)!}{L^{k+1}} \prod_{j=1}^{k+1} (g_j)$$

We must then apply the change of variables formula

$$P(s_1, \ldots, s_k) = P(\hat{s}_1, \ldots, \hat{s}_k) \times |J|$$

where $|J|$ is the Jacobian determinant of the transformation.

Since $s_j = L\hat{s}_j$, the inverse transformation is $\hat{s}_j = \frac{s_j}{L}$. Taking the derivative of each variable gives $\frac{\partial \hat{s}_j}{\partial s_j} = \frac{1}{L}$. The Jacobian determinant is then given by the product of the derivatives and since there are k points,

$$|J| = \left(\frac{1}{L}\right)^k = \frac{1}{L^k}$$

This allows us to obtain the final pdf

$$\pi(s_1, s_2, \ldots, s_k | M_k) = \frac{(2k+1)!}{L^{2k+1}} \prod_{j=1}^{k+1} g_j$$

$$\pi(s_1, s_2, \ldots, s_k | M_k) \propto \prod_{j=1}^{k+1} g_j$$

3. The constant rate model, M_0

The log likelihood of model M_0 is given by

$$\log \mathcal{L}(\{I_i\} | h_0, M_0) = \sum_{i=1}^{N} \log h_0 - \int_0^L dt\, x(t)$$

Since $x(t) = h_0$ is a constant over the interval L

$$\log \mathcal{L}(\{I_i\} | h_0, M_0) = N \log(h_0) - h_0 L$$

The likelihood is therefore

$$\mathcal{L}(\{I_i\} | h_0, M_0) = h_0^N \exp(-h_0 L)$$

From Exercise 2, the prior for the height parameters is universally given by a Gamma distribution $h_0 \sim \Gamma(\alpha, \beta)$ where $\alpha = 1$ and $\beta = 200$. This forms the probability density function

$$\pi(h_0 | M_0) = \frac{\beta^\alpha}{\Gamma(\alpha)} h_0^{\alpha-1} \exp(-\beta h_0)$$

Since $\alpha = 1$ and $\Gamma(1) = 1$, the prior simplifies to

$$\pi(h_0|M_0) = \beta \exp(-\beta h_0) = 200 \exp(-200 h_0)$$

Applying Bayes' rule the unnormalised posterior P^* is the product of the prior and likelihood

$$P^*(h_0|\{I_i\}, M_0) = \beta \cdot h_0^N \exp(-h_0(L + \beta))$$

To calculate the true evidence Z_0 for model M_0, we must integrate $P^*(h_0|\{I_i\}, M_0)$ over all possible values of h_0.

$$Z_0 = \int_0^\infty P^*(h_0|\{I_i\}, M_0)\, dh_0$$

When evaluating $P^*(h_0) = \beta \cdot h_0^N \exp(-h_0(L+\beta))$ for our dataset where $N = 191$ and $L = 40550$, the probabilities evaluate to 0 due to an effect known as floating point underflow.

To safely resolve this underflow, we perform the computation in log-space. First, we take the logarithm of the unnormalised posterior:

$$f(h_0) = \log(P^*(h_0)) = \log(\beta) + N \log(h_0) - h_0(L + \beta)$$

We can then find the exact maximum of this log-posterior analytically. Taking the derivative with respect to h_0 and setting to zero yields the maximum a posteriori (MAP) height:

$$h_{\max} = \frac{N}{L + \beta} \approx 0.00469$$

By evaluating the maximum log-posterior $C = f(h_{\max})$, we can shift the posterior to peak at 1 and prevent the underflow problem when calculating the evidence.

$$P^*(h_0) = \exp(C) \times \exp(f(h_0) - C)$$

$$Z_0 = \exp(C) \int_0^\infty \exp(f(h_0) - C)\, dh_0$$

We can now use numerical integration with `scipy.integrate.quad` to evaluate the integral

$$\log Z_0 = C + \log\left(\int_0^\infty \exp(f(h_0) - C)\, dh_0\right) \approx -1217$$

The prior and posterior distributions are plotted in Figure 3.

4. The 1 change point model, M_1

(a) MCMC Analysis

For the MCMC analysis, we use the `emcee` package which implements an affine invariant ensemble sampler. It works by creating an ensemble of multiple walkers that propose their next moves based on the position of the other walkers in the ensemble. This makes them invariant to the scale of the parameters, which is important given that s_1 is in the order of 10^4 and h_0,h_1 are in the order of 10^{-3}.

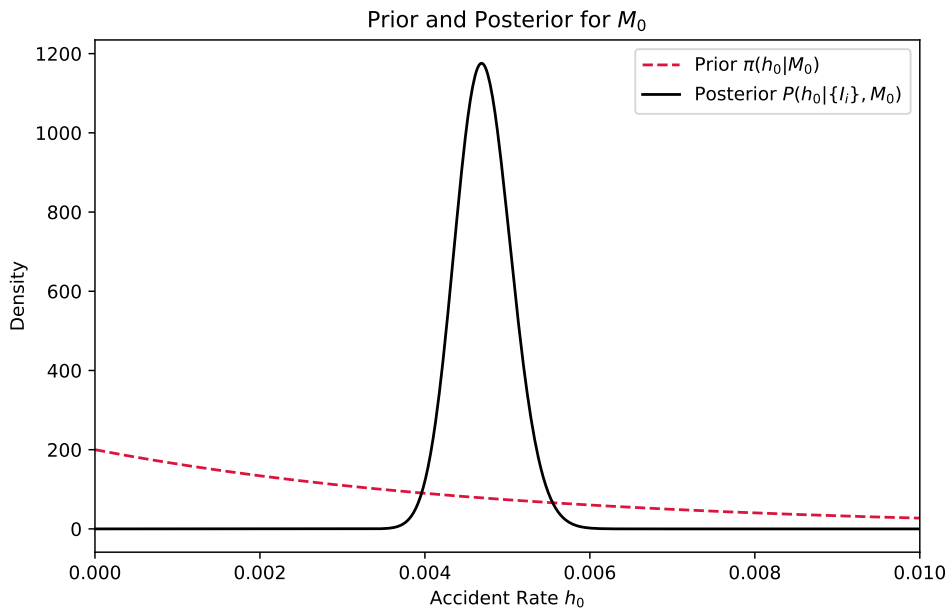

Figure 3: Prior (dashed line) and posterior (solid line) distributions for the constant rate model M_0. The maximum a posteriori value for the accident rate h_0 is approximately 0.0047 accidents per day.

There are 3 dimensions to search over $[s_1, h_0, h_1]$ so 32 walkers were used to ensure full coverage. From the trace plots in Figure 4 we can see that the chains converged quickly so only the first 500 steps were discarded as burn-in. This fast convergence is likely due to the fact that the walkers were initialised near plausible parameter values, chosen from the results of M_0. The trace plots each remain stable within a flat, horizontal band suggesting that they reached a stationary distribution. The autocorrelation times (τ) for the chains were calculated to be $44 < \tau < 52$ meaning that the 8000 steps were easily enough to obtain a large number of independent samples. The chains were not thinned as the sample size is too small to be concerned with memory usage. The Gelman-Rubin statistic was not calculated as the walkers look at other walkers to determine their next step, meaning they are inherently not independent.

The corner plot in Figure 5 shows that the s_1 parameter has 3 peaks and that there are several distinct change point locations that are plausible.

(b) Savage Dickey Density Ratio

The Savage-Dickey Density Ratio (SDDR) is a method for calculating the evidence ratio of two nested models with consistent priors. Since M_0 can be interpreted as a special case of M_1 where $h_0 = h_1$, the models are nested and the priors on the parameters h are consistent as they are given by the same Gamma distribution. This means that the SDDR can, in principle, be used to find the evidence ratio of M_1 to M_0.

We calculate the SDDR by the ratio of the posterior and prior density under the nested condition $\Delta h = h_0 - h_1 = 0$.

$$B_{01} = \frac{P(\Delta h = 0|D, M_1)}{\pi(\Delta h = 0|M_1)}$$

The prior density can be calculated analytically

$$\pi(\Delta h) = \frac{\beta}{2}\exp(-\beta|\Delta h|) = \frac{200}{2}\exp(0) = 100$$

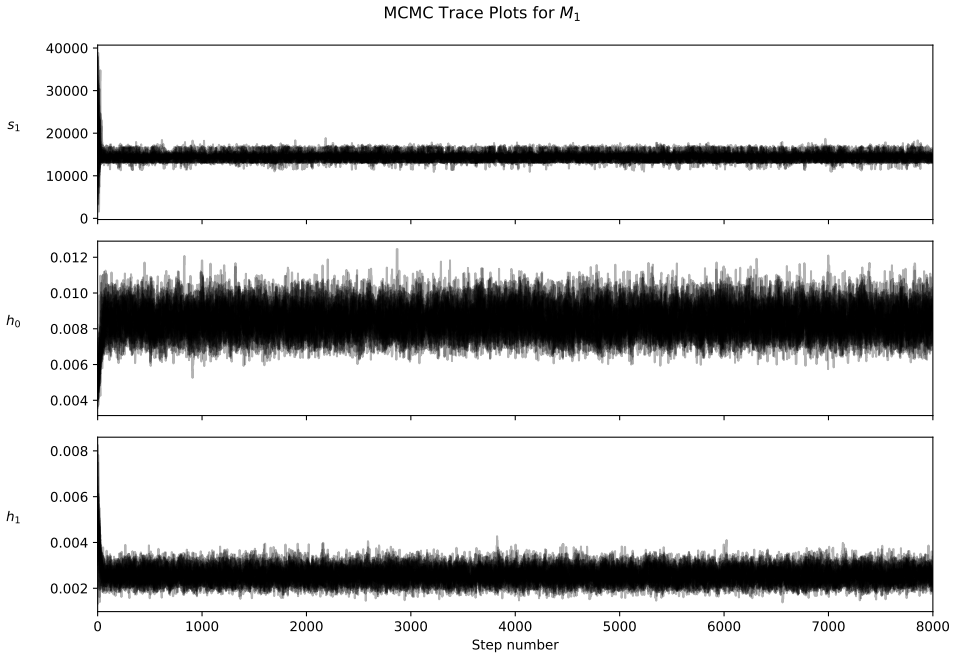

Figure 4: Trace plots from the ensemble MCMC sampler for the 1-change point model, M_1. The three panels show the evolution of the 32 walkers over 8000 steps for the parameters s_1, h_0, and h_1.

The problem arises in this example when trying to evaluate the posterior density at $\Delta h = 0$. From Figure 5, we can see that the marginalised posteriors for h_0 and h_1 are tightly constrained around 0.0085 and 0.0026 respectively, meaning there will be very little overlap and very few, if any, MCMC samples with $\Delta h = 0$. As a result, we can see that the kernel density estimate for the posterior difference in Figure 6 peaks at ≈ 0.006 and has effectively 0 density at $\Delta h = 0$. This makes the estimate for the SDDR highly numerically sensitive and unstable.

(c) Laplace Approximation

The Laplace approximation is a semi-analytic technique for calculating the evidence which approximates the posterior as a Gaussian around its peak. For M_1 there are two issues that prevent us from using this approximation effectively.

$$\log Z \approx \log P(D|\theta_{MAP}, M) + \log \pi(\theta_{MAP}|M) + \frac{d}{2}\log(2\pi) - \frac{1}{2}\log|H|$$

The Laplace approximation assumes the posterior can be well approximated by a Gaussian near its peak, so the log posterior must have well-defined local curvature. In this case, the condition is violated for the change point parameter s_1. As s_1 varies continuously, the log likelihood changes abruptly each time s_1 passes through an accident time y_i, because the rate assigned to that accident jumps between h_0 and h_1. This produces flat regions separated by abrupt jumps, so the log posterior is not locally quadratic around its maximum and the Hessian matrix H does not provide a meaningful description of the local curvature.

The second issue is that the posterior is multimodal, as can be seen in Figure 5 where the marginal posterior for s_1 has 3 distinct peaks. Fitting a single Gaussian around the MAP estimate would ignore these separated parts of probability mass, leading to an underestimation of the evidence.

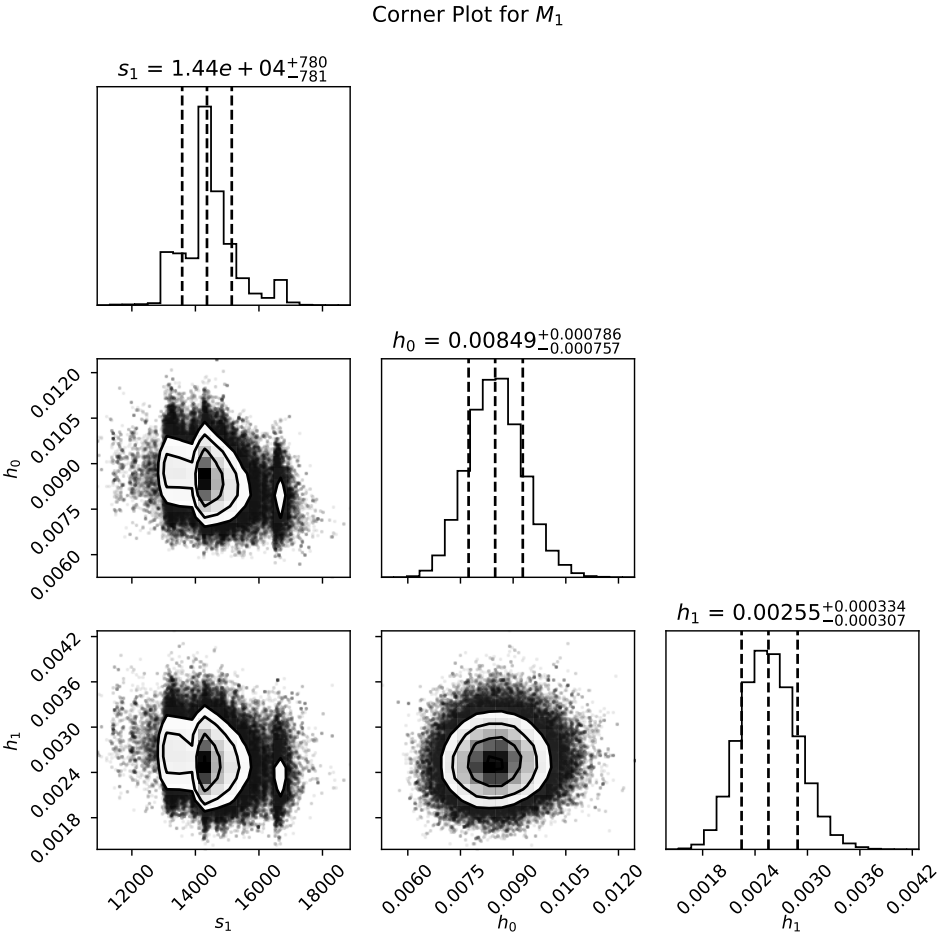

Figure 5: Corner plot showing the 1D and 2D marginal posterior distributions for the parameters s_1, h_0, and h_1 in the 1-change point model, M_1. The diagonal panels display the 1D marginal posteriors for each parameter, with dashed vertical lines indicating the 16th, 50th (median), and 84th percentiles.

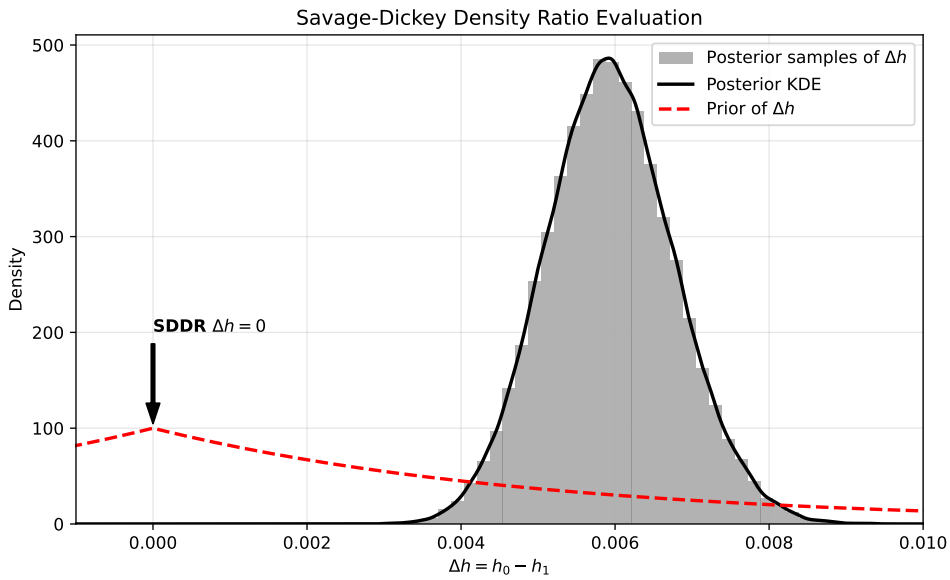

Figure 6: The plot displays the prior density (dashed red line), the posterior samples (grey histogram), and the posterior Kernel Density Estimate (solid black line) for the difference in accident rates, $\Delta h = h_0 - h_1$. The SDDR is evaluated at the nested condition $\Delta h = 0$, marked by the arrow. While the analytical prior density at this point is 100, the posterior density is effectively zero, with the distribution peaking around 0.006.

(d) Nested Sampling

To calculate the evidence Z we use the nested sampling package `dynesty`. Nested sampling works by initialising a set of live points and then iteratively replacing the points of lowest likelihood with one of higher likelihood. By keeping track of how the volume of the prior space changes the algorithm can numerically integrate the posterior to find Z.

We used 1000 live points to ensure a high resolution for numerical integration with a default stopping criterion of $\delta \log Z = 0.1$. Design choices to improve the efficiency of the algorithm included setting `bound='multi'` rather than the default single ellipsoid. The use of multiple bounds is important as we have seen in Figure 5 that the posterior is multimodal. Another choice was to use `sample='rwalk'` method for sampling, which is well suited to lower dimensional multimodal problems.

The calculated evidence for M_1 was $\log Z_1 = -1188.53 \pm 0.10$ resulting in a log Bayes factor $\log B_{10} \approx 28.56$ and, given the log prior odds $= \log 3$, the log posterior odds ≈ 29.66. This gives strong support for M_1 over M_0 under the chosen prior.

5. The k change point models, M_k

(b) MAP Change Points

The RJMCMC algorithm of following Green (1995) [1] was run for 300,000 iterations. The trace plot in Figure 7 shows that the chain explored a wide range of models and mixed well, staying mostly between $k = 2$ and $k = 5$, occasionally jumping as high as $k = 11$. The chain converged rapidly with the running mean stabilising within the first few thousand iterations, so only the first 10% of samples were discarded as burn-in. The posterior distribution on k peaks at $k = 3$ ($P \approx 0.28$), which is the maximum a posteriori estimate, followed by $k = 2$ ($P \approx 0.23$) and $k = 4$ ($P \approx 0.22$). The posterior is consistent with the Poisson prior, also peaked at $k = 3$, but

is more concentrated indicating that the data reinforces the prior belief without fully resolving model uncertainty.

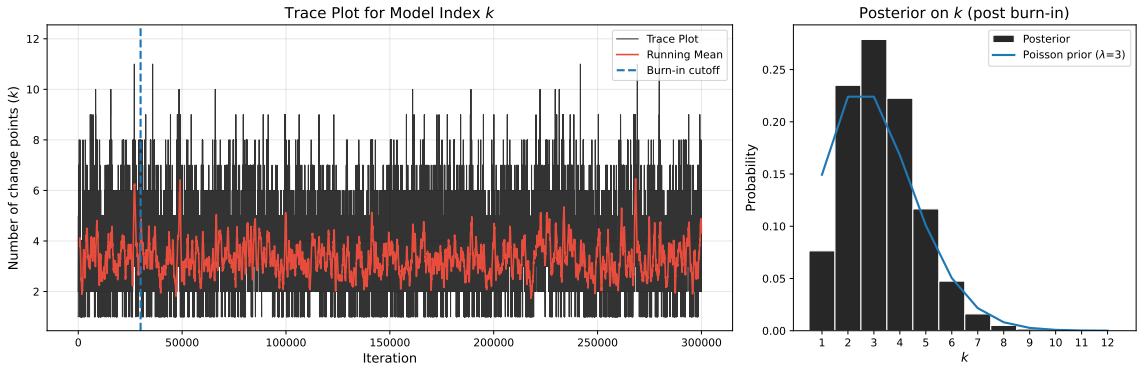

Figure 7: Reversible Jump MCMC (RJMCMC) results for the model index k over 300,000 iterations. **Left:** Trace plot of the number of change points, demonstrating that the chain successfully explores the model space up to $k = 11$. The red line shows the running mean ($n = 700$), which remains mostly between 2 and 5, indicating convergence. **Right:** The marginal posterior probability distribution for k with the Poisson prior plotted in blue. The posterior distribution peaks at $k = 3$, indicating the maximum a posteriori estimate.

(c) Model Averaging

After discarding the first 10% of samples as burn-in (based on the trace plot), the remaining RJMCMC samples were used for model averaging. The autocorrelation time was estimated to be $\tau \approx 195$, giving a moderate effective sample size of approximately 1384 so the estimates may be slightly noisy. The chain was not thinned as memory was not an issue.

To calculate the uncertainty regions, we define a grid of 500 evenly spaced time points over $[0, L]$. For each retained RJMCMC sample we evaluate $x(t)$ for each point in the grid, producing a $270,000 \times 500$ matrix where each row is an MCMC iteration and every column is a specific point in time.

Since RJMCMC samples models in proportion to their posterior, evaluating the distribution of $x(t)$ values at each time step (down the column) naturally marginalises over the number of change points k. We can then construct the median rate and the 50% and 90% credible intervals at each time point shown in Figure 8 by calculating the 5th, 25th, 50th, 75th, and 95th percentiles. Despite independently calculating the marginal uncertainty at each time point, the curve is smooth as it averages over many piecewise models. The final results suggest a strong decrease around 1890 with smaller variations around 1930 and 1945.

AI Use

I have used autogeneration tools to assist with code debugging, LaTeX formatting, helping to generate figures and creating ‘README.md‘. This declaration is made in accordance with the Course Handbook.

References

[1] P. J. Green, “Reversible jump Markov chain Monte Carlo computation and Bayesian model determination,” Biometrika, 82(4), 711-732, 1995.

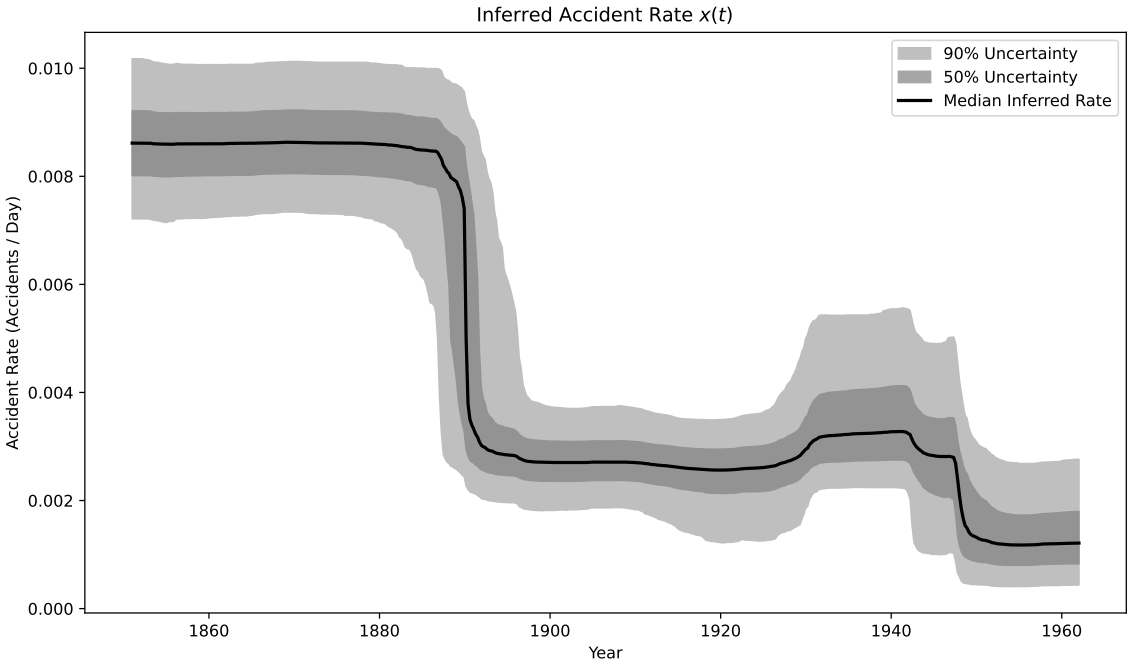

Figure 8: Inferred accident rate $x(t)$ over time, marginalised over the number of change points k using model averaging from the RJMCMC samples. The solid black line represents the median inferred rate. The dark and light grey shaded regions indicate the 50% and 90% credible intervals respectively, calculated from the percentiles of the posterior samples at each time step.